



S8: Introduction to Unsupervised Learning and Its Applications

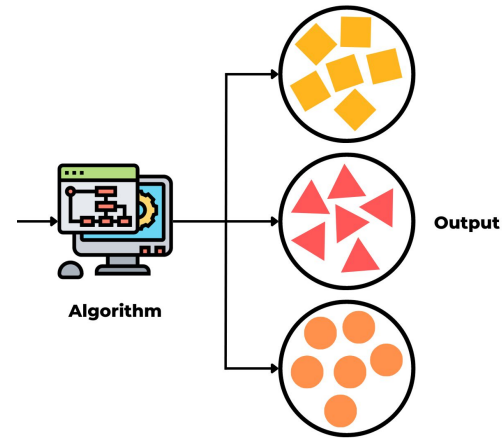
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Unsupervised Learning - Definition

Unsupervised learning in artificial intelligence is a type of machine learning that learns from data without human supervision.

- *Clustering*
- Anomaly Detection
- *Dimensionality Reduction*





Unsupervised Learning - *Simplified*

In unsupervised learning, models can identify **which data points are similar to each other**, but they do **not know the specific categories or classifications** of the data.

- **Clustering** = Groups similar data points together.
- Anomaly Detection = Identifies instances that **deviate from the usual patterns** in the data.



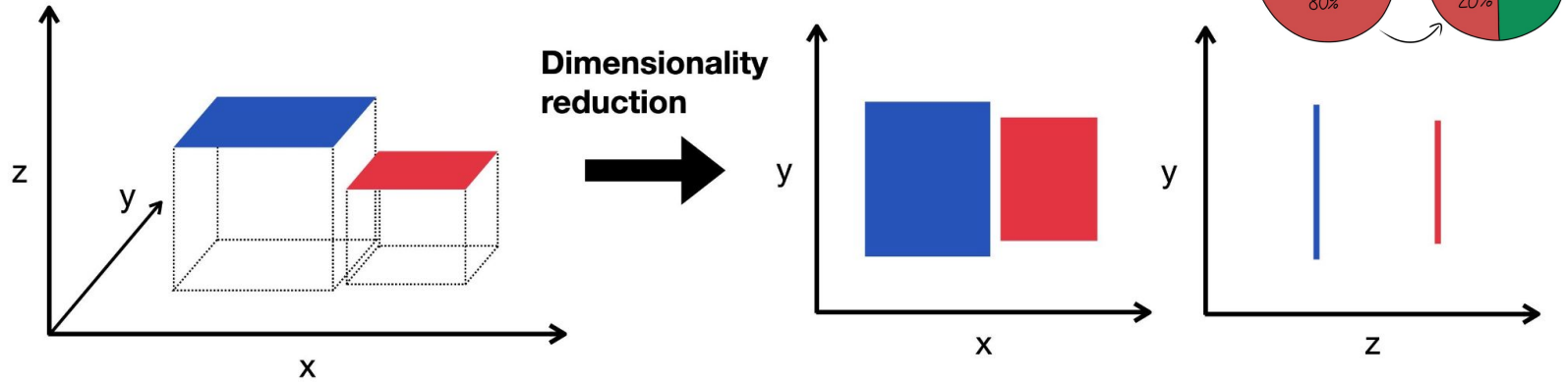
How About Dimensionality Reduction?

Dimensions refer to the **complexity** of a dataset, usually represented by the **number of independent variables (features)** it contains. **High-dimensional data can lead to overfitting!**

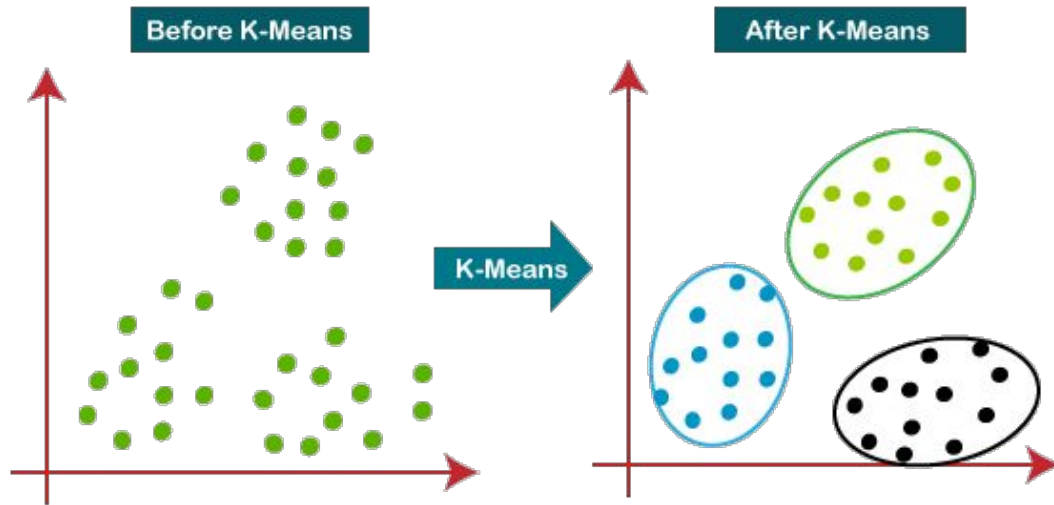
Dimensionality reduction is a technique used to reduce this complexity while preserving the most important information in the dataset, avoiding the **curse of dimensionality**.

Dimensionality Reduction - *Simplified*

Which features contribute the most to the patterns?

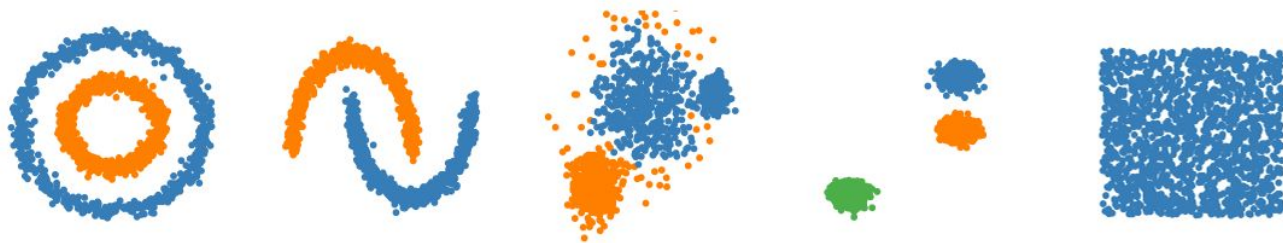


Centroid-Based Clustering - *K-Means*

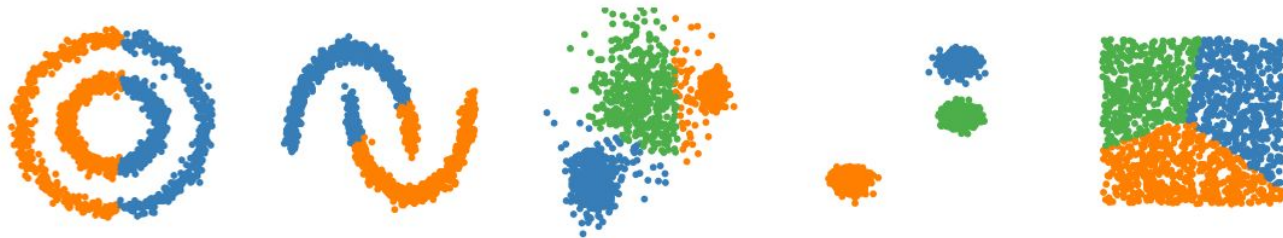


Density-Based Clustering - *DBSCAN*

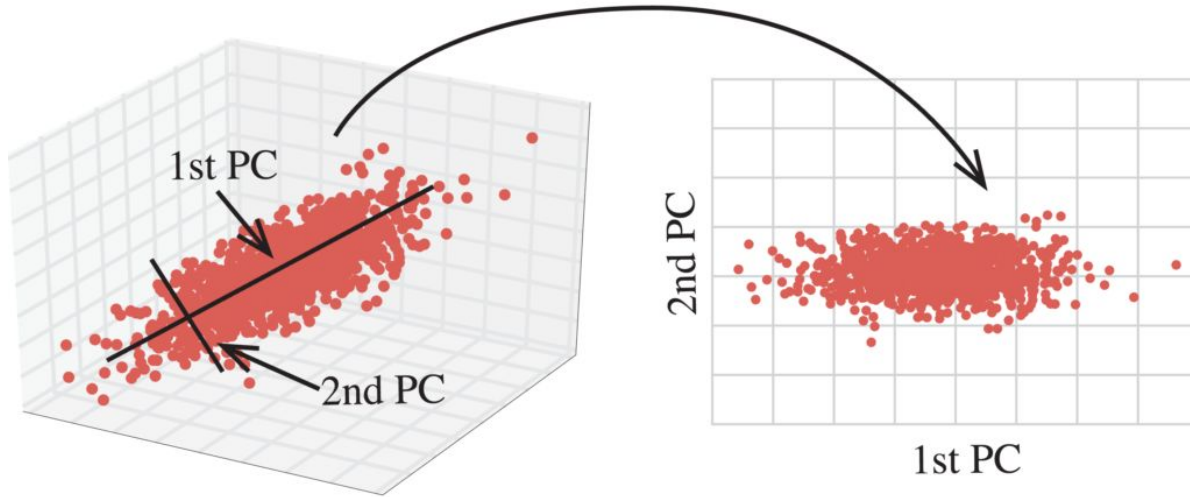
DBSCAN



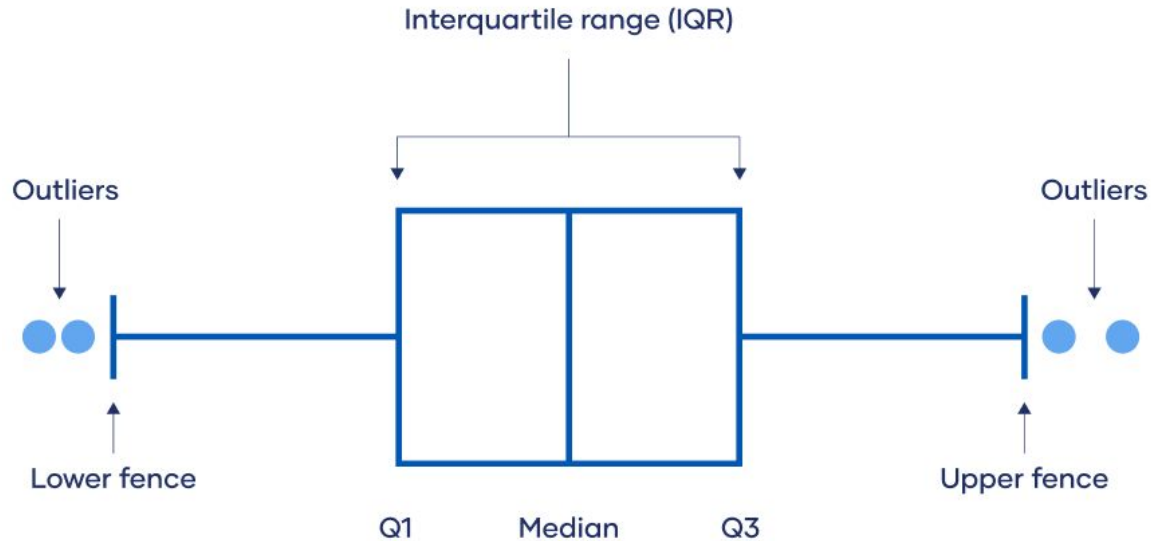
k-means



Principal Component Analysis (PCA)



Some Glances of Anomaly Detection - *Outliers*





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Learned in Action!**

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